26/03/2021

Code-A

Aakash
Medical|IIT-JEE|Foundations
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MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** Time : 3.00 Hrs.

Test - I5

Topics Covered : Complete Syllabus of Class-XI & XII (NEET Pattern)

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.
(ii) Mark should be dark and should completely fill the circle.
(iii) Dark only one circle for each entry.
(iv) Dark the circle in the space provided only.
(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.
(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

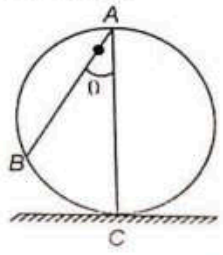
Choose the correct answer :

1. The contribution of S. Chandrasekhar to Physics is
   (1) Cosmic radiation
   (2) Nuclear Model & atom
   (3) LASER
   (4) Structure and evolution of star

2. The error in the measurement of the length of simple pendulum is 0.2% and the error in the time period is 3%. The possible percentage error in the physical quantity of dimensional formula [LT^{-2}] is
   (1) 4.2% (2) 6.2%
   (3) 3.2% (4) 4%

3. If a freely falling body covers half of its total distance in last second of its journey, then its time of fall will be
   (1) $\sqrt{2}$ s (2) 2 s
   (3) $\left(2+\sqrt{2}\right)$ s (4) $\left(1+\sqrt{2}\right)$ s

4. A frictionless wire is fixed between A and B inside a sphere of radius R as shown in figure. The time taken by the ball to slip from A to B is
   [Taking AC as diameter]



   (1) $\sqrt{\frac{R}{g}}$ (2) $\sqrt{\frac{R\cos\theta}{g}}$
   (3) $2\sqrt{\frac{R}{g}}$ (4) $\sqrt{\frac{2R\cos\theta}{g}}$

5. A body sliding down in a inclined plane have coefficient of friction 0.5. If the normal reaction is twice that of resultant downward force along incline. The angle between the inclined plane and the horizontal is
   (1) 45° (2) 60°
   (3) 30° (4) 37°

6. A body is revolving in a vertical circle of radius r with constant mechanical energy. The speed of the body at highest point is $\sqrt{3rg}$. The speed of the body at the lowest position will be

(1) $\sqrt{3gr}$ (2) $\sqrt{5gr}$

(3) $\sqrt{7gr}$ (4) $\sqrt{2gr}$

7. The coefficient of restitution depends upon

(1) The relative velocity of approach and separation

(2) The mass of colliding bodies

(3) The material of colliding bodies

(4) All of these

8. If $|\vec{A}| = 2$ N, $|\vec{B}| = 3$ N, then the value of $|\vec{A} \times \vec{B}|^2 + |\vec{A} \cdot \vec{B}|^2$ will be

(1) 144 units (2) 36 units

(3) 72 units (4) 90 units

9. A uniform meter rod is bent in L shape at its mid point at 90° to each other. The distance of centre of mass from the bent point will be

(1) $\frac{1}{4\sqrt{2}}$ m (2) $\frac{1}{2\sqrt{2}}$ m

(3) $\frac{1}{\sqrt{2}}$ m (4) $\frac{1}{8\sqrt{2}}$ m

10. For a body rolling along a level surface without slipping the translational and rotational kinetic energy are in the ratio 3 : 2. The body is

(1) Hollow sphere (2) Solid cylinder

(3) Hollow cylinder (4) Solid sphere

11. If amplitude of particle executing SHM is doubled, which of the following quantities are doubled?

a. Time period

b. Maximum velocity

c. Maximum acceleration

d. Total energy

(1) b and c (2) a, b and c

(3) a and c (4) a, b, c and d

12. A simple harmonic oscillator has time period of 4 s. The change in phase $\frac{3}{4}$ s after leaving the

(1) $\frac{3\pi}{4}$ (2) $\frac{3\pi}{8}$

(3) $\frac{\pi}{4}$ (4) $\frac{2\pi}{3}$

13. The orbital velocity of a satellite at a height $2R$ from the surface of the earth will be (R is radius of earth and 'g' is acceleration due to gravity.)

(1) $\sqrt{\frac{g}{R}}$ (2) $\sqrt{\frac{gR}{2}}$

(3) $\sqrt{gR}$ (4) $\sqrt{\frac{gR}{3}}$

14. Length of a string tied to two rigid supports is 60 cm. Maximum wavelength of stationary wave produced on it is

(1) 20 cm (2) 80 cm

(3) 40 cm (4) 120 cm

15. The frequency of 1st overtone of a closed pipe is 60 Hz. The frequency of 3rd overtone will be

(1) 140Hz (2) 120 Hz

(3) 80 Hz (4) 160 Hz

16. Water is coming out of a capillary tube drop by drop. If radius of capillary tube is r and weight of drop is W, then W is equal to

(1) πrT (2) $\pi r^2 T$

(3) $4\pi rT$ (4) $2\pi rT$

17. A material has normal density ρ and bulk modulus K. The increase in the density of the material when it is subjected to an external pressure P from all sides is

(1) $\frac{P}{\rho K}$ (2) $\frac{KP}{\rho}$

(3) $\frac{P\rho}{K}$ (4) $\frac{\rho K}{P}$

18. The coefficient of volume expansion of water is negative between

(1) 0 K and 4 K (2) 273 K and 277 K

(3) 300 K and 304 K (4) 0°C to 10°C

19. Two gases A and B having same pressure P, volume V and absolute temperature T are mixed. If mixture have volume and temperature as V and T respectively, then the pressure of the mixture is

(1) $2P$ (2) P

(3) $\frac{P}{2}$ (4) $4P$

20. The power radiated by a black body at temperature 200 K is 544 W. Its surface area is $\left(\sigma = 5.67\times10^{-8}\,Wm^{-2}K^{-4}\right)$

(1) $6\times10^{-2}m^2$ (2) $6\ m^2$

(3) $6\times10^{-6}m^2$ (4) $6\times10^{2}m^2$

21. The temperature of the system decreases in the process of

(1) Free expansion

(2) Adiabatic expansion

(3) Isothermal expansion

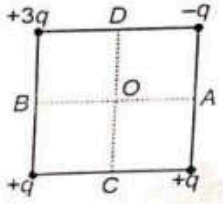
(4) Isothermal compression

22. Initially, sphere A has charge $-60e$ and sphere B has charge of $+20e$. The sphere are made of conducting material and are identical in size. If spheres touch each other, the resulting charge on sphere A will be

(1) $-15e$ (2) $-20e$

(3) $+25e$ (4) $+30e$

23. Four charges are arranged at the corners of a square as shown in figure. The direction of electric field at the centre of the square is along



(1) DC (2) BC

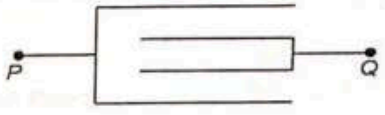
(3) BA (4) AD

24. An electric dipole of dipole moment P is aligned parallel to a uniform electric field E. The energy required to rotate the dipole by 180° is

(1) PE (2) $\frac{PE}{2}$

(3) $2PE$ (4) Zero

25. Four identical plates each of area A are separated by a distance d as shown in the figure. The capacitance between point P and Q will be



(1) $\frac{A\varepsilon_0}{2d}$ (2) $\frac{4A\varepsilon_0}{d}$

(3) $\frac{2A\varepsilon_0}{d}$ (4) $\frac{A\varepsilon_0}{d}$

26. Fractional increase in resistivity per unit increase in temperature is defined as

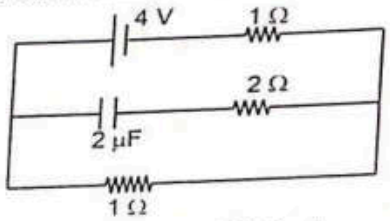
(1) Resistivity

(2) Temperature coefficient of resistivity

(3) Conductivity

(4) Drift velocity

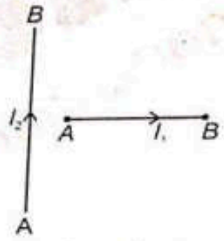
27. The energy stored in 2 μF capacitor as shown in the circuit in steady state will be



(1) 4 μJ (2) 2 μJ

(3) 0.5 μJ (4) 1 μJ

28. A current I_1 carrying wire AB is placed near another long wire CD carrying current I_2. If wire AB is free to move, it will have



(1) Rotational motion only

(2) Translational motion only

(3) Rotational as well as translational motion

(4) Neither rotational nor translational motion

29. A small square loop of wire of each side l is kept inside a large square loop of wire each side 'L' ($L >> l$). The loops are coplanar and their centres coincide. The mutual inductance of the system is proportional to (current i is in outer loop)

(1) $\frac{l}{L}$ (2) $\frac{l^2}{L}$

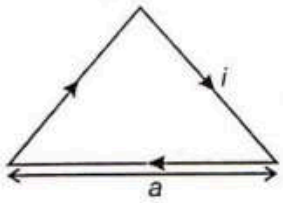
(3) $\frac{L}{l}$ (4) $\frac{L^2}{l}$

30. When orientation of dipoles parallel and antiparallel to magnetic field is distributed unequally, then the material is

(1) Paramagnetic (2) Ferromagnetic

(3) Ferrimagnetic (4) Antiferromagnetic

31. The magnetic field at the centroid of a current carrying equilateral triangular wire of side length a as shown in the figure will be



(1) $\frac{6\mu_0 i}{\pi a}$ (2) $\frac{9\mu_0 i}{2\pi a}$

(3) $\frac{3\mu_0 i}{2\pi a}$ (4) $\frac{\mu_0 i}{\pi a}$

32. At a frequency ω_0, the reactance of a certain capacitor equal that of certain inductor. If frequency changed to $2\omega_0$, the ratio of reactance of inductor to that of the capacitor is

(1) 4 : 1 (2) $\sqrt{2}$: 1

(3) $1 : 2\sqrt{2}$ (4) 1 : 2

33. Maxwell's equation describe the fundamental laws of

(1) Electricity only

(2) Magnetism only

(3) Mechanics only

(4) Both (1) and (2)

34. If focal length of a double convex lens for red light is f_r, its focal length for the violet light is

(1) f_r

(2) Greater than f_r

(3) Less than f_r

(4) $2 f_r$

35. At what distance from a convex mirror of focal length 2 m should a boy stand so that his image has height equal to half the original height?

(1) 2 m from the mirror

(2) 1 m from the mirror

(3) 4 m from the mirror

(4) 0.5 m from the mirror

36. In Young's double slit experiment the fringe width 8 mm. If the experiment shifted is to water of refractive index $\frac{4}{3}$, the fringe width becomes (in mm)

(1) 6 (2) 4

(3) 10 (4) 8

37. Light of wavelength 6000 Å is incident on a single slit. The first minimum of diffraction pattern is obtained at 4 mm from the centre. The screen is at a distance of 2 m from the slit. The slit width will be

(1) 0.3 mm

(2) 0.2 mm

(3) 0.15 mm

(4) 0.1 mm

38. If the kinetic energy of free electron is doubled, its de-Broglie wavelength changes by a factor

(1) 2 (2) $\frac{1}{2}$

(3) $\sqrt{2}$ (4) $\frac{1}{\sqrt{2}}$

39. The velocity of the most energetic electron is doubled when the frequency ν of incident radiation is doubled. The work function of the metal is

(1) $\frac{h\nu}{4}$ (2) $\frac{h\nu}{3}$

(3) $\frac{h\nu}{2}$ (4) $\frac{2h\nu}{3}$

40. In Bohr model of hydrogen atom, which of the following is quantized?

(1) Linear momentum of electron

(2) Linear velocity of electron

(3) Angular momentum of electron

(4) Angular velocity of electron

41. The ratio of magnetic moment to the orbital angular momentum for an electron is

(1) 8.5×10^{-10} C/kg

(2) 6.7×10^{-10} C/kg

(3) $8.8 \times 10^{+10}$ C/kg

(4) $6.8 \times 10^{+10}$ C/kg

42. An alpha nucleus of energy 1/2 mv^2 bombards a heavy nuclear target of charge Ze. Then the minimum distance of closest approach is proportional to

(1) $\frac{1}{Ze}$ (2) v^2

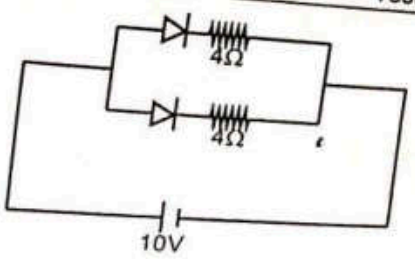
(3) $\frac{1}{m}$ (4) $\frac{1}{v^4}$

43. When a selective impurity is doped into an intrinsic semiconductor, the conductivity of semiconductor

(1) Increases (2) Decreases

(3) Remains same (4) Becomes zero

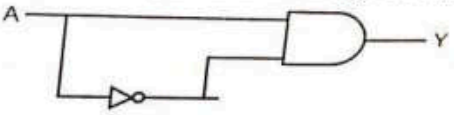
44. The current flowing in through the battery as shown in figure, is [Assuming diodes are ideal]



(1) 5 A (2) 2 A

(3) 2.5 A (4) Zero

45. The Boolean expression for the given circuit is



(1) $Y = A$ (2) $Y = A \cdot \overline{A}$

(3) $Y = 1$ (4) $Y = A + \overline{A}$

CHEMISTRY

46. Which among the following gas has minimum critical temperature?

(1) H_2 (2) He

(3) O_2 (4) N_2

47. In which of the following reactions entropy increases?

(1) $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

(2) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

(3) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

(4) $H_2O(g) \rightarrow H_2O(l)$

48. Which of the following mixture does not act as an acidic buffer?

(1) $H_2SO_4 + SO_4^{2-}$ (2) $H_2CO_3 + HCO_3^-$

(3) $HPO_4^{2-} + PO_4^{3-}$ (4) $H_2PO_4^- + HPO_4^{2-}$

49. Oxidation state of S in $H_2S_2O_8$ and H_2SO_5 respectively are

(1) +4, +6 (2) +6, +5

(3) +6, +4 (4) +6, +6

50. The essential amino acid among the following is

(1) Serine (2) Arginine

(3) Proline (4) Alanine

51. Neoprene contains

(1) Cl atoms (2) S atoms

(3) N atoms (4) O atoms

52. Select the incorrect statement among the following

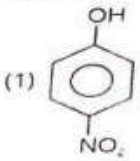
(1) Mixture of chloroxylenol and terpineol acts as antiseptic

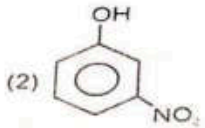
(2) Equanil is employed as a tranquilizer

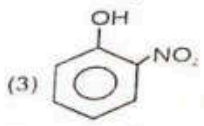
(3) Non-ionic detergents are esters

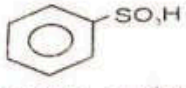
(4) Alitame is 600 times sweeter than cane sugar

53. Which among the following is least acidic in nature?

(1) 

(2) 

(3) 

(4) 

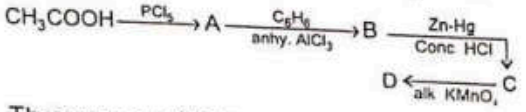
54. Most reactive compound towards nucleophilic addition out of the following is

(1) Ethanal (2) Propanone

(3) Methanal (4) Benzaldehyde

55. Which of the following gives aldol condensation reaction?
(1) Chloral
(2) Benzophenone
(3) Methyl ethyl ketone
(4) 2, 2-dimethylpropanal

56. Consider the following reaction sequence



The compound D is
(1) Benzoic acid (2) Benzyl alcohol
(3) Phenol (4) Phenyl acetic acid

57. The correct order of basic strength of ethyl substituted amines in aqueous solution is
(1) $(C_2H_5)_3N > (C_2H_5)_2NH > C_2H_5NH_2$
(2) $(C_2H_5)_2NH > C_2H_5NH_2 > (C_2H_5)_3N$
(3) $(C_2H_5)_3N > C_2H_5NH_2 > (C_2H_5)_2NH$
(4) $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2$

58. Electrolytic reduction of nitrobenzene in weakly acidic medium and strongly acidic medium respectively produces
(1) Aniline, azoxybenzene
(2) Aniline, p-aminophenol
(3) Azobenzene, p-aminophenol
(4) Azoxybenzne, azobenzene

59. Which of the following is a mineral of copper?
(1) Siderite
(2) Malachite
(3) Cryolite
(4) Sphalerite

60. Incorrect order among the following is
(1) Boiling point : $SbH_3 > NH_3 > AsH_3$
(2) Bond energy : $F_2 > Cl_2 > Br_2$
(3) Acidic strength : $HClO_4 > HClO_3 > HClO_2$
(4) Thermal stability : $H_2S > H_2Se > H_2Te$

61. In which of the following ions *d-d* transition takes place?
(1) $Cr_2O_7^{2-}$ (2) MnO_4^{2-}
(3) CrO_4^{2-} (4) MnO_4^-

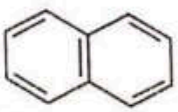
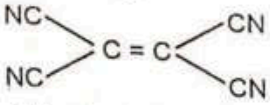
62. P–H bond is absent in
(1) $H_4P_2O_5$ (2) $H_4P_2O_6$
(3) H_3PO_3 (4) H_3PO_2

63. The electrode potentials for Cu^{2+}/Cu^+ and Cu^{2+}/Cu are +0.15 V and +0.32 V respectively. The electrode potential for Cu^+/Cu will be
(1) + 0.49 V
(2) +0.17 V
(3) +0.02 V
(4) +0.56 V

64. For the strong electrolytes NaCl, NaOH and $BaCl_2$; the molar conductivities at infinite dilution (in S m^2 mol^{-1}) are x, y and z respectively. The molar conductivity of $Ba(OH)_2$ at infinite dilution (in S m^2 mol^{-1}) will be
(1) z – 2y + 2x
(2) 2z – x + y
(3) z – 2x + 2y
(4) z – 2x – 2y

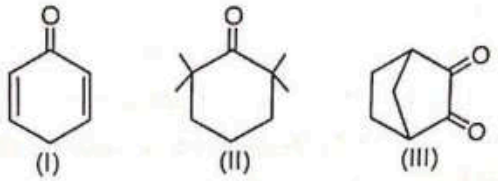
65. For the reaction, $A(g) + 2B(g) \rightarrow 4C(g) + 3D(g)$
The rate of reaction can be expressed by
(1) $\frac{4d[C]}{dt}$ (2) $-\frac{1}{3}\frac{d[D]}{dt}$
(3) $\frac{d[A]}{dt}$ (4) $-\frac{1}{2}\frac{d[B]}{dt}$

66. Negatively charged sol among the following is
(1) $CrO_3.xH_2O$ sol
(2) Methylene blue sol
(3) TiO_2 sol
(4) Sb_2S_3 sol

67. In which of the following molecules,all atoms are not coplanar?
(1) 
(2) 
(3) 

(4) $CH \equiv C - CH = CH_2$

68. Which among the given molecules can exhibit tautomerism?



(I) (II) (III)

(1) Both (I) and (III)
(2) III only
(3) I only
(4) Both (II) and (III)

69. Some o, p-directing substituents in aromatic electrophilic substitution are given. Which one is excepted to be most activating?

(1) –Cl (2) –OH
(3) $-CH_3$ (4) $-NHCOCH_3$

70. An alkene (A) on reductive ozonolysis gives propanone and methanal in equimolar ratio. Addition of HBr to alkene (A) gives (B) as the major product. The name of (B) is

(1) 1-bromo-2-methylpropane
(2) 2-bromobutane
(3) 2-bromo-2-methylpropane
(4) 1-bromobutane

71. Mole fraction of solvent in aqueous solution of 10 % (w/w) NaOH solution is

(1) 0.95 (2) 0.84
(3) 0.91 (4) 0.72

72. Percentage of Mg in a biomolecule is 0.2% by weight. The minimum molecular weight of the biomolecule is

(1) 1.2×10^5 (2) 2.4×10^3
(3) 2.4×10^5 (4) 1.2×10^4

73. The orbital angular momentum of electron in 3*p* orbital is

(1) Zero (2) $\frac{h}{2\pi}$
(3) $\frac{2h}{\pi}$ (4) $\frac{h}{\sqrt{2}\pi}$

74. The de-Broglie wavelength of a particle having mass 1 mg and moving with velocity 1000 m/s is

(1) 6.63×10^{-34} m (2) 6.63×10^{-31} m
(3) 6.63×10^{-35} m (4) 6.63×10^{-33} m

75. Tolerable limit of nitrates and sulphates in drinking water respectively are

(1) 500 ppb, 50 ppb (2) 500 ppm, 50 ppb
(3) 50 ppm, 500 ppm (4) 50 ppm, 500 ppb

76. The incorrect order against the indicated property is

(1) Metallic character : Mg < Si < Na
(2) 1st ionization energy : B < Be < C
(3) Ionic radii : $Na^+ < F^- < O^{2-}$
(4) Electron affinity : Br < F < Cl

77. Hydrogen bond is not observed in

(1) HCl (2) H_2O
(3) HF (4) NH_3

78. Shape of I_3^+ and I_3^- respectively are

(1) Bent, linear (2) Linear, linear
(3) Bent, Bent (4) Linear, Bent

79. In which species only π-bond is present?

(1) O_2 (2) C_2
(3) N_2 (4) F_2

80. Consider the thermal dissociation of solid A_2B_3

$A_2B_3(s) \rightleftharpoons 2AB(g) + B(g)$

If total pressure at equilibrium is 3 atm then K_P of the reaction will be

(1) $1\ atm^2$ (2) $3\ atm^2$
(3) $9\ atm^3$ (4) $4\ atm^3$

81. In a solid compound having cubic unit cells, the atoms A are present at corners of cubes, B atoms are present at body centre and atoms C are present at face centres and edge centres. The formula of the compound is

(1) ABC_3 (2) A_2BC
(3) ABC_6 (4) AB_2C

82. Which among the following aqueous solutions has highest boiling point?

(1) 0.1 M urea (2) 0.1 M NaCl
(3) 0.1 M Na_2SO_4 (4) 0.1 M NH_4NO_3

83. What is the correct electronic configuration of the central metal atom in $[CoF_6]^{3-}$ based on crystal field theory?

(1) $t_{2g}^6\ e_g^0$ (2) $t_{2g}^4\ e_g^2$
(3) $t_{2g}^3\ e_g^3$ (4) $t_{2g}^5\ e_g^1$

84. Which of the following species is optically active?

(1) $[Co(en)_3]Cl_3$

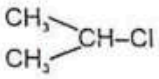
(2) $[Ni(NH_3)_4(H_2O)_2]Cl_2$

(3) trans – $[Co(en)_2Cl_2]Cl$

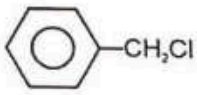
(4) $[Co(NH_3)_3Cl_3]$

85. Among the following most reactive compound towards S_N1 reaction is

(1) $CH_2 = CH - CH_2 - Cl$

(2) 

(3) $CH_2 = CH - Cl$

(4) 

86. Heating benzyl-phenyl ether with one equivalent HI produces

(1) Iodo benzene (2) Benzyl alcohol

(3) Carbolic acid (4) 4-Iodo phenol

87. Industrially H_2O_2 is prepared by

(1) The action of aq. H_2SO_4 on hydrated BaO_2

(2) Electrolytic oxidation of acidified sulphate solutions followed by hydrolysis

(3) The auto-oxidation of 2-ethylanthraquinol

(4) By passing CO_2 through a cold pasty solution of BaO_2

88. The chemical reagent used to remove the hardness of water in Clark's method is

(1) Na_2CO_3 (2) $Na_6P_6O_{18}$

(3) $NaAlSiO_4$ (4) $Ca(OH)_2$

89. Which of the following is required as cofactor for enzymes that utilise ATP in phosphate transfer?

(1) Ca (2) Mg

(3) Na (4) K

90. Which of the following is mostly readily soluble in water?

(1) $MgSO_4$ (2) $BaSO_4$

(3) $SrSO_4$ (4) $CaSO_4$

BOTANY

91. Select the **correct** statements w.r.t organelles included in 'Endomembrane system'.

(1) Bring about the oxidation of carbohydrates, proteins and β-oxidation of fats

(2) Performs storage of starch

(3) Contains water, sap, excretory products and other material not useful for the cell

(4) Is of spherical shape and found in the nucleoplasm

92. Which of the given seal the ends of chromosomes and prevent their shortening or chromosome loss?

(1) Telomere (2) Kinetochore

(3) Primary constriction (4) Centromere

93. Select the **correct** set of events which occurs in G_2 phase of cell cycle?

(A) Duplication of semi-autonomous cell organelles.

(B) Duplication of centrioles.

(C) Synthesis of tubulin protein.

(D) DNA replication.

(1) Both (A) and (B) (2) Both (A) and (C)

(3) Both (B) and (C) (4) Both (C) and (D)

94. Match the **Column I** with **Column-II** and select the **correct** option.

	Column-I		Column-II
a.	Leptotene	(i)	Homologous chromosomes clearly appears as tetrad
b.	Zygotene	(ii)	Terminalisation of chiasmata
c.	Pachytene	(iii)	Begins with condensation and coiling of chromatin fibres.
d.	Diakinesis	(iv)	Activity of synaptonemal complex

(1) a(iii), b(i), c(ii), d(iv) (2) a(i), b(iv), c(iii), d(ii)

(3) a(ii), b(iii), c(i), d(iv) (4) a(iii), b(iv), c(i), d(ii)

95. Select the **incorrect** match w.r.t taxonomic categories of Brinjal

(1) Genus – Solanum
(2) Order – Polymoniales
(3) Class – Monocotyledonae
(4) Division – Angiospermae

96. Match the Column-I with Column-II and select the **correct** option.

	Column-I		Column-II
a	Key	(i)	Collection of preserved plants and animals
b	Monograph	(ii)	Identification of plants and animals based on similarities and dissimilarities
c	Botanical garden	(iii)	Contains information of any one taxon
d	Museum	(iv)	*Ex-situ* conservation strategies of plants

(1) a(ii), b(iii), c(iv), d(i) (2) a(ii), b(iv), c(iii), d(i)
(3) a(iv), b(iii), c(i), d(ii) (4) a(iii), b(i), c(iv), d(ii)

97. Hyphal walls contains cellulose and other glucans in which of the given fungi?

(1) *Phytophthora* (2) *Rhizopus*
(3) *Aspergillus* (4) *Neurospora*

98. How many kingdoms w.r.t Whittaker's kingdom classification system exhibit both autotrophic as well as heterotrophic mode of nutrition?

(1) 1 (2) 2
(3) 4 (4) 3

99. The arrangement of flowers on the floral axis of the plant is called

(1) Aestivation (2) Inflorescence
(3) Placentation (4) Bolting

100. The scar on the seed coat through which developing seeds were attached to fruit is called

(1) Hilum (2) Micropyle
(3) Testa (4) Tegmen

101. In dicot leaf, vascular bundles differ in size. It is due to

(1) Absence of bulliform cells
(2) Distribution of more stomata on lower epidermis
(3) Presence of reticulate venation
(4) Dorsiventral type of leaf

102. Conjoint closed vascular bundles is seen in

(1) Dicot root (2) Monocot root
(3) Dicot stem (4) Monocot stem

103. Which of the given algae is a rich source of iodine?

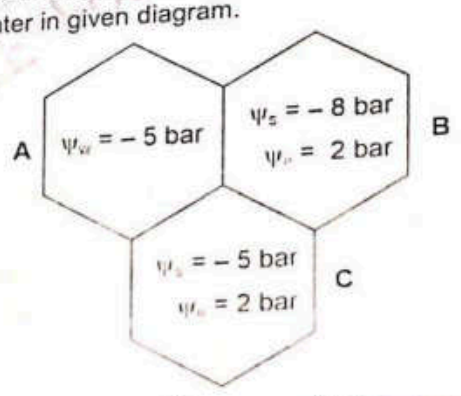
(1) *Gelidium*
(2) *Porphyra*
(3) *Chara*
(4) *Laminaria*

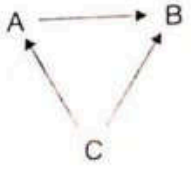
104. How many of the given plants listed in box have archegonia as well as produce seeds in their life-cycle?

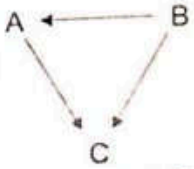
Chara, Pinus, Cedrus, Lycopodium, Marsilea, Ginkgo, Mustard, Wolffia.

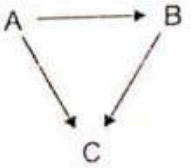
(1) 5 (2) 3
(3) 4 (4) 1

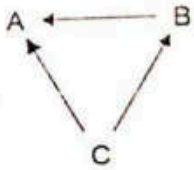
105. Select the **correct** option w.r.t direction of flow of water in given diagram.



(1) 

(2) 

(3) 

(4) 

106. The process of absorption of water by hydrophilic solid particles of a substance without forming a solution is called
(1) Imbibition
(2) Plasmolysis
(3) Guttation
(4) Translocation

107. In which of the given region of roots, there is initiation of nodule formation takes place?
(1) Root hair (2) Pericycle
(3) Inner cortex (4) Epidermis

108. Which of the given bacteria exhibits autotrophic mode of nutrition and fixes nitrogen symbiotically as well as in free-living state?
(1) *Rhizobium* (2) *Frankia*
(3) *Anabaena* (4) *Azotobacter*

109. Select the **incorrect** statement w.r.t C_3 pathway.
(1) The primary acceptor of CO_2 is a three carbon compound
(2) Occurs in the mesophyll cells of leaves
(3) 3 ATP are consumed to fix one CO_2
(4) 2 NADPH are required to fix one molecule of carbon dioxide

110. Which of the given is not **true** w.r.t. photosystem II (or, PS-II)?
(1) It occurs on inner surface of thylakoids
(2) It is found in both grana and stroma lamella
(3) It is involved only in non-cyclic flow of electrons
(4) It is associated with splitting of water and release of O_2

111. Which of the given complexes in ETS contains cytochromes a and a_3 and two copper centre?
(1) NADH dehydrogenase
(2) Complex-II
(3) Cytochrome c oxidase complex
(4) Cytochroma bc_1 complex

112. Which of the given respiratory substrate has least respiratory quotient (RQ) under aerobic respiration?
(1) Glucose (2) Tripalmitin
(3) Malic acid (4) Oxalic acid

113. Select the **incorrect** match w.r.t phytohormones and their composition
(1) Indole compounds – Auxin
(2) Terpenes – Ethylene
(3) Adenine derivatives – Kinetin
(4) Derivatives of carotenoids – ABA

114. How many of the given plants in the box do not show any correlation between exposure to light duration and induction of flowering response?

Tomato, Wheat, Radish, Tobacco, Cucumber, Soyabean

(1) 4 (2) 3
(3) 5 (4) 2

115. Select the **incorrect** statement w.r.t most of the species of *Chara*
(1) Have male sex organ called globule
(2) Have female sex organ called nucule
(3) Are monoecious
(4) Globule occupies an upper position than nucule

116. Zoospores and zygote produced in algae differs from each other as the former
(1) Is always diploid
(2) Is non-motile
(3) Is a sexual reproductive structure
(4) Is usually naked

117. Which of the given plants are **not** pollinated by water?
(1) *Hydrilla* (2) *Vallisneria*
(3) *Zostera* (4) Water hyacinth

118. Pollen grains are well-preserved as fossils because of the presence of
(1) Sporopollenin
(2) Cellulose
(3) Pectin
(4) Chitin

119. How many types of gametes can be produced by a diploid organism, if it is heterozygous for three loci and homozygous for two loci?
(1) 8 (2) 32
(3) 3 (4) 5

120. Select the **incorrect** statement w.r.t phenylketonuria (PKU)
(1) It is an inborn error of metabolism
(2) It is inherited as autosomal recessive trait
(3) The affected individuals lack a liver enzyme phenylalanine hydroxylase that converts tyrosine into phenylalanine
(4) The affected individuals are mentally retarded

121. All of the given are stop codons, **except**
(1) AUG
(2) UAA
(3) UAG
(4) UGA

122. Which of the given chromosomes have maximum and least genes respectively w.r.t human genome?
(1) Chromosome 1 and X
(2) Chromosome X and Y
(3) Chromosome Y and 1
(4) Chromosome 1 and Y

123. Pusa snowball K-1 bred by hybridisation and selection for resistance to black rot and curl blight black rot is a variety of which of the given crop?
(1) Cowpea
(2) Cauliflower
(3) Chilli
(4) *Brassica*

124. Recovery of healthy plants from diseased plants is possible by
(1) Meristem culture
(2) Somatic hybridisation
(3) Protoplast fusion
(4) Leaf culture

125. All of the given are fermented products prepared by using bacteria, **except**
(1) Curd
(2) Toddy
(3) Dosa
(4) Idli

126. Match the column-I and column-II and select the **correct** option.

	Column-I		Column-II
a	Streptokinase	(i)	Blood-cholesterol lowering agent
b	Cyclosporin A	(ii)	Clot buster
c	Statins	(iii)	Detergent formulations
d	Lipase	(iv)	Immunosuppressive agent

(1) a(i), b(ii), c(iii), d(iv)
(2) a(iv), b(iii), c(ii), d(i)
(3) a(ii), b(iii), c(i), d(iv)
(4) a(ii), b(iv), c(i), d(iii)

127. The basic unit of study in ecology is
(1) Organism
(2) Ecosystem
(3) Biome
(4) Community

128. Which of the given contributes to an increase in population density?
(1) Natality and Immigration
(2) Mortality and Emigration
(3) Immigration and Emigration
(4) Mortality and Immigration

129. All of the given are most productive ecosystems, **except**
(1) Coral reefs
(2) Tropical rain forest
(3) Sugarcane field
(4) Deep sea

130. In which of the given areas, secondary succession occurs?
(1) Abandoned farm lands
(2) Sand dunes
(3) Cooled volcanic lava
(4) Igneous rock

131. Select the **Incorrect** match w.r.t biodiversity in Amazonian rain forest (including taxa and number of species).
(1) Mammals – 427
(2) Birds – 1300
(3) Reptiles – 3000
(4) Amphibian – 427

132. Which of the following is an example of organism with recent extinction in Africa?
(1) Thylacine (2) Quagga
(3) Dodo (4) Steller's sea cow

133. Colloidal material present as impurities in waste water is
(1) Sand (2) Faecal matter
(3) Silt (4) Ammonia

134. Joint forest management was started by Government of India in
(1) 1980 (2) 1966
(3) 1974 (4) 1986

135. Which of the given is **not** associated with C_3 cycle?
(1) Carboxylation
(2) Reduction
(3) Regeneration of RuBP
(4) Release of oxygen

ZOOLOGY

136. Which of the following animals evolve only upto tissue level of body organisation?
(1) *Scypha* (2) *Ancylostoma*
(3) *Hydra* (4) *Bombyx*

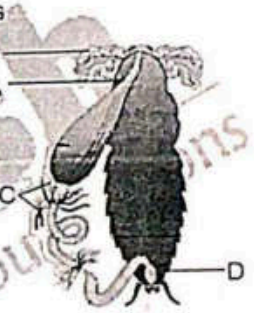
137. Complete the analogy
Platyhelminths : Triploblastic acoelomate :: Aschelminths : _____
(1) Triploblastic coelomate
(2) Diploblastic coelomate
(3) Triploblastic pseudocoelomate
(4) Radially symmetrical, coelomate

138. All of the following are features of molluscs **except**
(1) Organ-system level of body organisation
(2) Metameric segmentation
(3) Development may be direct or indirect
(4) They are mostly dioecious

139. Presence of which of the following is most unique mammalian characteristics?
(1) Mammary glands
(2) Pulmonary respiration
(3) Viviparity
(4) Four chambered heart

140. Select the **correct** match w.r.t animals and their common names
(1) *Panthera leo* – Tiger
(2) *Ornithorhyncus* – Flying fox
(3) *Scoliodon* – Dog fish
(4) *Icthyophis* – Toad

141. In the following given figure, identify A, B, C and D. Choose the option which **correctly** represents their features



	A	B	C	D
(1)	Brain-response to stimuli	Pharynx-used for storing food	Hepatic caecae-secretes digestive juice	Rectum-exchange of gases
(2)	Reservoir-grinding of food	Crop-storage of food	Hepatic caecae-excretion of uric acid	Rectum-secretion of uric acid
(3)	Oesophagus-passage for food	Salivary gland-secretes saliva	Hepatic caecae - secretes digestive juice	Rectum-opens out through anus
(4)	Salivary reservoir-storage of saliva	Gizzard-grinding of food particles	Hepatic caecae-secretes digestive juice	Rectum-excretory in function

142. Read the following statements A and B and choose the option with **incorrect** answer.

Statement-A: On an average, female of cockroach produces 14–16 ootheca, each containing 9–10 eggs.

Statement-B: The nymph grows by moulting about 10 times to reach the adult cockroach.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

143. All of the following are secretions of endocrine glands **except**

(1) Insulin

(2) Adrenaline

(3) Milk

(4) Glucagon

144. Select the **correct** match

(1) Epithelial tissue — Cells are loosely arranged with large intercellular matrix

(2) Connective tissue — Most abundant and widely distributed tissue in the body of complex animals

(3) Muscular tissue — Generally play an active role in all the movements of only GIT

(4) Neural tissue — Exerts the greatest control over the body's responsiveness to changing conditions only in surroundings

145. Match the enzymes in Column-I with their respective substrate in column-II and choose the option with all **correct** match.

	Column-I		Column-II
a.	Lipase	(i)	Dipeptides
b.	Nuclease	(ii)	Fats
c.	Carboxypeptidase	(iii)	Nucleic acids
d.	Dipeptidase	(iv)	Proteins and its derivatives

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iii), b(ii), c(i), d(iv)

146. Read the following statements A and B and choose the **correct** answer.

Statement-A : Paneth cells are present in crypts of Lieberkühn and secrete lysozymes to kill bacteria.

Statement-B : The site, smell and/or the presence of food in the oral cavity can stimulate the secretion of saliva, gastric juice and succus entericus.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

147. Select the **incorrect** match.

(1) Jaundice — Liver is affected, skin and eyes turn yellow

(2) Vomiting — Increased liquidity of faecal discharge

(3) Constipation — Faeces are retained in colon

(4) Indigestion — Caused due to inadequate enzyme secretion, anxiety, over eating and spicy food

148. Which of the following is **not** the function of conducting part of respiratory system?

(1) Gaseous transport

(2) Humidify inspired air

(3) Clears inspired air from foreign particles

(4) Gaseous exchange

149. Select the approximate amount of air left in lungs after maximal forced expiration in a normal woman

(1) 0.1 L

(2) 1.1 L

(3) 1.6 L

(4) 2.8 L

150. Which of the following factors is responsible for dissociation of oxyhaemoglobin?

(1) High pH

(2) High pCO_2

(3) Low H^+ concentration

(4) Low temperature

151. In normal human blood

(1) Eosinophils are the most abundant type of blood cells

(2) There are more lymphocytes than neutrophils

(3) Iron is mostly trapped in haemoglobin

(4) There are more platelets than red blood cells

152. Opening of which of the following voltage gated channels contribute to the repolarisation phase of skeletal muscle fibres?

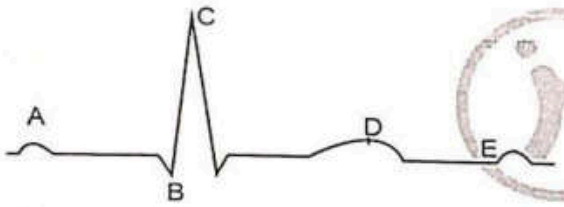
(1) Na^+ channels

(2) Ca^{+2} channels

(3) Cl^- channels

(4) K^+ channels

153. Diagrammatic presentation of a standard ECG is given below?



How many of the labels in above ECG identify the part of ECG with events during cardiac cycle **correctly**?

(i) A-represents the atrial depolarisation

(ii) D-represents ventricular repolarisation

(iii) E-represents the end of joint diastole to start atrial systole

(iv) C-corresponds to maximum opening of ventricular sodium channels

(v) B-represents S-wave of standard ECG

(1) 2

(2) 3

(3) 4

(4) 5

154. In the presence of vasopressin, the greatest fraction of filtered water is reabsorbed in the

(1) Proximal tubule

(2) Loop of Henle

(3) Distal tubule

(4) Medullary interstitium

155. Read the following statements carefully and choose the option with only **correct** statements.

a. In fishes, ammonia is generally excreted by diffusion through gill surfaces as ammonium ions

b. Protonephridia in *Planaria* is primarily concerned with osmoregulation

c. Kidneys are situated close to dorsal inner wall of the abdominal cavity

d. On an average 550–600 ml of blood is filtered by each kidney per minute

(1) a, b and c only (2) b, c and d only

(3) a, b, c and d (4) c and d only

156. Select the **incorrect** match

(1) Fibrous joints – Between cranial bones

(2) Ball and socket joint – Between humerus and pelvic girdle

(3) Pivot joint – Between atlas and axis vertebra

(4) Gliding joint – Between the carpals

157. Match columns A and B and choose the option with all **correct** match

	Column-A		Column-B
a.	Thin filament	(I)	Formed by head and short arm of myosin monomer
b.	Troponin	(ii)	Monomeric protein constitute one thick filament
c.	Cross arm	(iii)	Distributed at regular intervals on the tropomyosin
d.	Meromyosin	(iv)	Formed by 2 filaments of F-actin, 2 filaments of tropomyosin which run close to the 'F' actins through out its length

(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(ii), d(i)

(3) a(i), b(iv), c(ii), d(iii) (4) a(iv), b(iii), c(i), d(ii)

158. Autonomic nerve fibres which are unmyelinated, characterised by all of the following features **except**

(1) Presence of nurilemma

(2) Presence of Schwann cells

(3) Less conduction velocity than myelinated nerve fibres

(4) Presence of nodes of Ranvier

159. Receptors stimulated in knee-jerk reflex are present in

(1) Sarcolemma of muscle

(2) Skin

(3) Muscle spindle

(4) Patella

160. In organ of Corti, the basal end of hair cells are in close contact with the A and above the rows of hair cells is a thin elastic membrane called B. Choose the option which fill the blanks A and B **correctly**

	A	B
1.	Efferent nerve fibres	Tectorial membrane
2.	Afferent nerve fibres	Basilar membrane
3.	Motor nerve fibres	Reissner's membrane
4.	Afferent nerve fibres	Tectorial membrane

161. Choose the **odd** one w.r.t diseases associated with hypersecretion of endocrine glands

(1) Graves' disease

(2) Addison's disease

(3) Gigantism

(4) Acromegaly

162. All of the following hormones directly or indirectly helps in erythopoiesis **except**

(1) Thyroxine

(2) Erythropoietin

(3) Cortisol

(4) Oestrogen

163. Which one of the following is **not** an application of transgenic animals?

(1) Some transgenic animals serve as transgenic models for human diseases to study the treatment

(2) First transgenic cow, Rosie produced human protein-enriched milk

(3) α-1-antitrypsin obtained from transgenic sheep, Tracy is used in treatment of emphysema

(4) Transgenic chicken has been successfully developed to cure the Hepatitis-B

164. Due to ampicillin resistance gene, one is able to select a transformed cell in the presence of ampicillin. The ampicillin resistance gene in this case is called

(1) Recombinant (2) Selectable marker

(3) Origin of replication (4) Recognition site

165. *Taq* polymerase used during PCR is actually a/an

(1) Thermostable DNA polymerase

(2) Thermolabile DNA polymerase

(3) RNA directed DNA polymerase

(4) DNA directed RNA polymerase

166. Super-ovulation and embryo transfer are meant for improving

(1) Human race (2) Livestock

(3) Poultry (4) Fishery

167. A person without thymus gland would **not** be able to

(1) Produce antibodies by B-lymphocytes

(2) Reject a tissue transplant

(3) Receive a tissue transplant

(4) Produce anti allergic response

168. Which of the following techniques uses non-ionizing radiations to accurately detect cancer of internal organs?

(1) X-rays

(2) Computed tomography

(3) MRI

(4) Immunotherapy

169. Diacetylmorphine is commonly known as

(1) Smack (2) Coke

(3) Hashish (4) Morphine

170. Select the **mismatch** w.r.t. innate immunity in normal adult human

(1) Physical barriers — Epithelial lining of urinogenital tract without mucus

(2) Physiological barriers — Saliva in the mouth

(3) Cellular barriers — Monocytes

(4) Cytokine barriers — Interferons

171. Read the following statements carefully and choose the option with only **incorrect** statements.

a. Immuno-suppressants are not required after organ transplantation.

b. IgA antibodies present in colostrum provide natural active immunity to the infants.

c. The use of drugs like histamine, adrenaline and cortisol quickly reduce symptoms of allergy.

d. Primary and secondary immune response is carried out only by B-lymphocytes.

(1) a, b and c only (2) a, b and d only

(3) b, c and d only (4) a, b, c and d

172. Who discarded the theory of spontaneous generation forever?

(1) Louis Pasteur (2) Von Baer

(3) Charles Darwin (4) Stanley Miller

173. Separate the following examples into homologous and analogous organs

a. Sweet potato and potato

b. Flippers of penguins and dolphins

c. Hearts of different vertebrates

d. Forelimbs of whales, bats and cheetah

e. Thorns of *Bougainvillea* and tendrils of *Cucurbita*

f. Eyes of *Octopus* and humans

g. Mouthparts of insects

	Homologous organs	Analogous organs
1.	a, b, c	d, e, f, g
2.	c, d, e, g	a, b, f
3.	a, b, c, d	e, f, g
4.	b, c, d	a, e, f, g

174. Human babies with average weight at the time of birth have more chances of survival than heavy weight or low weight new borns. It illustrates

(1) Stabilizing selection

(2) Gene flow

(3) Diversifying selection

(4) Founder effect

175. Choose the **odd** set w.r.t ploidy in following given options

(1) Spermatogonia, oogonia, meiocyte

(2) Meiocyte, primary spermatocyte, primary oocyte

(3) Spermatid, spermatozoa, secondary oocyte

(4) Ootid, spermatid, first polar body, primary oocyte

176. Cell division is itself a mode of reproduction in

(1) *Amoeba*

(2) *Hydra*

(3) *Planaria*

(4) *Taenia*

177. Read the following statements A and B and choose the **correct** answer.

A : The presence of X or Y chromosome in the sperm determines the sex of embryo.

B : Mammary glands differentiate during pregnancy and starts producing milk towards end of pregnancy.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

178. Select the **incorrect** match w.r.t hormone, their site of action and effect

	Hormone	Site of Action	Effect
(1)	FSH	Sertoli cell	Spermatogenesis
(2)	LH	Interstitial cells	Secrete androgens
(3)	Oestrogen	Mammary glands	Lactation
(4)	Oxytocin	Uterus	parturition

179. Which of following is a bacterial STI?

(1) Trichomoniasis

(2) AIDS

(3) Genital warts

(4) Syphilis

180. Which of the following co-factor forms coordinate bonds with side chains at the active site and at the same time form one or more coordinate bonds with the substrate?

(1) Prosthetic groups (2) Haem

(3) Coenzymes (4) Metal ions

❑ ❑ ❑

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